



Monitoring and Alerting — TASCO Growth Platform

TASCO Insurance × VETC Growth Platform — Operations

iorta TechNXT

Version 1.0 · 07 October 2026

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Item	Value
Owner	SRE (iorta TechNXT during hypercare → TASCO IT operations)
Stack (reference)	Prometheus (+ Alertmanager), Grafana, Loki (or the TASCO log platform), blackbox exporter, postgres_exporter / sql_exporter. Any Prometheus-compatible stack works
Related	Runbook (RB-xx) · Performance plan §4 SLOs · KI-14, KI-17

1. What the application exposes

Endpoint	Content	Exposure
GET /metrics	Prometheus text format 0.0.4, per process : counters and histograms below, plus process_resident_memory_bytes and process_uptime_seconds	Protected by METRICS_TOKEN when set: Prometheus scrapes with authorization: {type: Bearer, credentials_file: ...}. Open when unset (KI-14) : always set it, block /metrics at the ingress, and let the NetworkPolicy allow only the Prometheus namespace
GET /health/live	{status:"ok"}	Liveness probe
GET /health/ready	{status, store, db}; 503 when not ready	Readiness probe; external synthetic
GET /api/ops/status	Circuit state per integration, active rule versions and checksums, eventBacklog by status, audit count	Staff with ops: read. Use for humans and dashboards' links, not as a scrape target
stdout JSON logs	request lines with requestId, method, route, status, ms; lifecycle and error events	Log pipeline (Loki/ELK)

Notes on the metrics implementation (src/shared/metrics.js):

- Counters are in-memory per pod and **reset on restart**. Always use rate() / increase() and sum across pods.
- The output has no # TYPE lines, so Prometheus ingests the series as untyped. rate() and histogram_quantile() still work on the _bucket series.
- Histogram buckets (seconds): 0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10, +Inf.
- route is the **route template** (e.g. POST /api/customer/orders, GET /api/customers/:id). Probes are health, the scrape itself is metrics, unmatched/static paths are static, and the spec is openapi, so cardinality is bounded (73 API routes + 4).

2. Additional exporters (to configure)

Some signals the app does not export yet (KI-17). Provide them with a read-only SQL exporter against the **replica** (role metrics_ro), polled every 60 s. These series are prefixed tasco_db_ to make clear they come from the database, not the app.

Series	Query (sketch)	Used by
tasco_db_domain_events{status}	SELECT status, count(*) FROM domain_events GROUP BY status	ALR-21, ALR-22
tasco_db_domain_events_oldest_pending_seconds	SELECT extract(epoch FROM now() - min(occurred_at)) FROM domain_events WHERE status='pending'	ALR-21
tasco_db_touchpoints_due_unexecuted	SELECT count(*) FROM touchpoints WHERE status='scheduled' AND due_date < CURRENT_DATE	ALR-34
tasco_db_handoffs_open_older_than_2h	SELECT count(*) FROM handoffs WHERE status='open' AND created_at < now() - interval '2 hours'	ALR-83
tasco_db_orders{status} (last 24 h)	SELECT status, count(*) FROM orders WHERE created_at > now()-interval '1 day' GROUP BY status	ALR-84
tasco_db_job_last_success_timestamp{kind}	SELECT kind, extract(epoch FROM max(started_at)) FROM job_runs WHERE data->'status'='succeeded' GROUP BY kind	ALR-35
postgres_exporter standard	Connections, replication lag, locks, transaction wraparound, DB size	ALR-90–ALR-93

Only reconciliation and retention record job_runs today. CLI-run journeys, recompute and relay do not, so their freshness comes from logs (ALR-33) and the CronJob status (kube_cronjob_status_last_successful_time from kube-state-metrics).

3. Metric catalogue (from the code)

Metric	Type	Labels	Emitted by
http_requests_total	counter	route, method, status	app.js, every request
http_request_duration_seconds	histogram	route, method	app.js
integration_calls_total	counter	integration (vetc-wallet, tasco-core, voice-ai, app-push, zalo-zns, sms), result (ok, error)	resilience.js, every attempt including retries
integration_short_circuit_total	counter	integration	resilience.js, calls rejected while the circuit is open
events_published_total	counter	type (policy.issued, profiles.rebuilt, lead.recompute_requested, handoff.created, renewal.link_requested, claim.submitted, rules.activated)	outboxEventBus.publish
events_processed_total	counter	type, status (done, pending = will retry, dead_letter)	outboxEventBus.relay
messages_total	counter	channel (app_push, zalo_zns, sms), status (sent, failed, blocked), journey	journeyService.sendMessage
voice_calls_total	counter	outcome (hot_handoff, link_sent, already_renewed, opted_out, wrong_person, plate_mismatch, unverified, callback_later, not_interested)	voiceService.finalize

Metric	Type	Labels	Emitted by
quotes_total	counter	channel (vetc_app , zalo , telesales , voice_bot , partner_api)	salesService.quote
orders_completed_total	counter	channel , journey	salesService.purchase
process_resident_memory_bytes	gauge	—	metrics.render
process_uptime_seconds	gauge	—	metrics.render

4. SLIs and SLOs

SLO ID	User journey	SLI (Prometheus)	Objective (28-day rolling)
SLO-01	Customer purchase path availability	Non-5xx ratio over route=~"POST /api/customer/(quotes orders) POST /api/customer/session GET /api/customer/home"	99.9 % (40 min error budget / 28 d)
SLO-02	Partner API availability	Non-5xx ratio over route=~"(POST GET) /api/partner/v1/.*"	99.9 %
SLO-03	Staff console availability	Non-5xx ratio over all other /api/* routes	99.5 %
SLO-04	Certificate verification availability	Non-5xx ratio over GET /api/public/certificates/:certNo + synthetic SYN-02	99.95 %
SLO-05	Latency: customer reads	p95 of GET /api/customer/home	≤ 300 ms for 99 % of 5-min windows
SLO-06	Latency: quotes	p95 of quote routes	≤ 500 ms
SLO-07	Latency: orders	p95 of order routes	≤ 2.5 s
SLO-08	Latency: staff reads	p95 of GET /api/leads , GET /api/customers/:id , GET /api/handoffs	≤ 500 ms
SLO-09	Journey freshness	Daily journey run completes by 12:00 ICT with backlog tasco_db_touchpoints_due_unexecuted = 0	99 % of days
SLO-10	Event freshness	Oldest pending event < 5 min	99 % of the time

Recording rules:

```

groups:
- name: tasco-gp-recording
  rules:
  - record: route:http_requests:rate5m
    expr: sum by (route, status) (rate(http_requests_total[5m]))
  - record: slo:customer_purchase:error_ratio5m
    expr: |
      sum(rate(http_requests_total{route=~"POST /api/customer/(quotes|orders)|POST
/api/customer/session|GET /api/customer/home",status=~"5.."}[5m]))
      /
      sum(rate(http_requests_total{route=~"POST /api/customer/(quotes|orders)|POST
/api/customer/session|GET /api/customer/home"}[5m]))
  - record: slo:customer_purchase:error_ratio1h
    expr: |
      sum(rate(http_requests_total{route=~"POST /api/customer/(quotes|orders)|POST
/api/customer/session|GET /api/customer/home",status=~"5.."}[1h]))
      /
      sum(rate(http_requests_total{route=~"POST /api/customer/(quotes|orders)|POST
/api/customer/session|GET /api/customer/home"}[1h]))
  - record: route:http_request_duration_seconds:p95_5m
    expr: histogram_quantile(0.95, sum by (le, route)
(rate(http_request_duration_seconds_bucket{route!~"health|metrics|static|openapi"}[5m])))
  - record: integration:error_ratio5m
    expr: |
      sum by (integration) (rate(integration_calls_total{result="error"}[5m]))
      / sum by (integration) (rate(integration_calls_total[5m]))

```

5. Golden signals and dashboards

Dashboard	Panels	Audience
D1 Service overview	RPS by route group; 4xx/5xx ratio; p50/p95/p99 by route (route:http_request_duration_seconds:p95_5m); 429 rate; pods ready; memory per pod; restarts; SLO burn (SLO-01..04)	On-call
D2 Integrations	Per integration: call rate, error ratio, short-circuits, circuit state (from integration_short_circuit_total > 0 plus a link to /api/ops/status), retries (calls ÷ requests)	On-call, vendor managers
D3 Events and jobs	events_published_total vs events_processed_total{status} by type; tasco_db_domain_events{status} ; oldest pending; dead letters; CronJob last success; journey run summary (log-derived)	On-call, L3
D4 Business funnel	quotes_total and orders_completed_total by channel and journey; quote → order conversion; messages by channel and status; voice outcomes; open handoffs > 2 h; GWP (from the /api/dashboard/overview export, daily)	Business owner, campaign managers
D5 Compliance	messages_total{status="blocked"} ; marketing messages outside the window (SQL, should be 0); opt-outs (voice_calls_total{outcome="opted_out"}); DNC contacts (should be 0); audit chain verify status; rule activations timeline	Compliance, DPO
D6 Security	Login 401/423/429 rates; MFA failures (audit); 403 rate by route; partner 401 rate; WAF blocks	Security
D7 Database	Connections vs max, CPU, IOPS, replication lag, slow statements, table sizes (domain_events , audit_log , messages), advisory-lock waits	DBA

Golden signals mapping:

Signal	Metric
Latency	<code>http_request_duration_seconds</code> (by route), <code>ms</code> in logs
Traffic	<code>rate(http_requests_total[5m])</code> , <code>quotes_total</code> , <code>orders_completed_total</code>
Errors	<code>http_requests_total{status=~"5.."} , integration_calls_total{result="error"} , events_processed_total{status="dead_letter"} , messages_total{status="failed"}</code>
Saturation	<code>process_resident_memory_bytes</code> , container CPU throttling, DB connections, <code>tasco_db_domain_events{status="pending"}</code>

6. Alert rules

`severity: page` pages on-call 24×7; `severity: ticket` creates a ticket and posts to the channel during business hours. Every alert carries `runbook_url` to the RB-xx section.

6.1 Platform

ID	Name	Expression	For	Severity	Runbook
ALR-01	ReadinessFailing	<code>probe_success{job="blackbox",instance=~".*/health/ready"} == 0</code>	2m	page	RB-01
ALR-02	HighErrorRate	<code>sum(rate(http_requests_total{status=~"5..",route!~"health metrics static"}[5m])) / sum(rate(http_requests_total{route!~"health metrics static"}[5m])) > 0.02</code>	5m	page	RB-01 / RB-12
ALR-03	LatencySLOBreach	<code>route:http_request_duration_seconds:p95_5m{route=~"GET /api/customer/home"} > 0.3</code> or <code>route:http_request_duration_seconds:p95_5m{route=~"POST /api/(customer/)?quotes POST /api/partner/v1/quotes"} > 0.5</code> or <code>route:http_request_duration_seconds:p95_5m{route=~"POST /api/(customer/ partner/v1/)?orders"} > 2.5</code>	10m	ticket (page if > 2× threshold)	RB-12
ALR-04	PurchaseSLOFastBurn	<code>slo:customer_purchase:error_ratio5m > (14.4 * 0.001)</code> and <code>slo:customer_purchase:error_ratio1h > (14.4 * 0.001)</code>	2m	page	RB-02/03
ALR-05	PurchaseSLOSlowBurn	<code>slo:customer_purchase:error_ratio1h > (6 * 0.001)</code>	30m	ticket	—
ALR-06	InternalErrors	<code>sum(increase(http_requests_total{status="500"}[10m])) > 5</code> (500 = INTERNAL_ERROR ; 503 is upstream)	—	ticket	Logs unhandled error
ALR-07	RateLimitSurge	<code>sum(rate(http_requests_total{status="429"}[5m])) > 5</code>	10m	ticket	RB-09 / RB-14
ALR-08	NoTraffic	<code>sum(rate(http_requests_total{route!~"health metrics static"}[15m])) == 0</code> and <code>on() (hour() >= 1 and hour() < 13) (08:00–20:00 ICT)</code>	15m	page	RB-01

ID	Name	Expression	For	Severity	Runbook
ALR-09	ScrapedOwn	<code>up{job="tasco-growth-platform"} == 0</code>	5m	ticket	—

6.2 Integrations

ID	Name	Expression	For	Severity	Runbook
ALR-10	CircuitOpen	<code>sum by (integration) (increase(integration_short_circuit_total[5m])) > 0</code>	1m	page for vetc-wallet, tasco-core; ticket for others	RB-02 – RB-05
ALR-11	IntegrationErrorRatio	<code>integration:error_ratio5m > 0.2 and sum by (integration) (rate(integration_calls_total[5m])) > 0.1</code>	5m	ticket	RB-02 – RB-05
ALR-12	IntegrationRetryInflation	<code>sum by (integration) (rate(integration_calls_total[10m])) / on() group_left sum(rate(http_requests_total{route=~"POST /api/(customer/ partner/v1/)?orders"}[10m])) > 3</code> (wallet: > 1 call per order means retries)	15m	ticket	RB-02

6.3 Events and jobs

ID	Name	Expression	For	Severity	Runbook
ALR-20	DeadLetterEvents	<code>sum by (type) (increase(events_processed_total{status="dead_letter"}[15m])) > 0</code>	—	page (business hours) / ticket (night)	RB-06
ALR-21	OutboxBacklog	<code>tasco_db_domain_events{status="pending"} > 10000 or tasco_db_domain_events_oldest_pending_seconds > 300</code>	10m	ticket (page if > 30 min)	RB-06
ALR-22	EventsStuckProcessing	<code>tasco_db_domain_events{status="processing"} > 0</code> sustained. The Postgres relay re-claims after 5 min, so persistence means no relay is running	15m	ticket	RB-06
ALR-23	EventRetryChurn	<code>sum by (type) (rate(events_processed_total{status="pending"}[10m])) > 0.5</code>	15m	ticket	RB-06
ALR-24	BacklogGrowth (app-only fallback)	<code>sum(increase(events_published_total[30m])) - sum(increase(events_processed_total{status="done"}[30m])) > 5000</code>	30m	ticket	RB-06
ALR-33	JourneySpike (log-based)	Loki ruler: <code>max_over_time({app="tasco-growth-platform"} json msg="journey run complete" unwrap skipped [1h]) / max_over_time({app="tasco-growth-platform"} json msg="journey run complete" unwrap due [1h]) > 0.3</code>	—	ticket	RB-07

ID	Name	Expression	For	Severity	Runbook
ALR-34	JourneyBacklog	<code>tasco_db_touchpoints_due_unexecuted > 0 and on() hour() >= 5 (after 12:00 ICT)</code>	30m	ticket	RB-07 / KI-05
ALR-35	JobNotRun	<code>time() - kube_cronjob_status_last_successful_time{cronjob=~".*(journeys reconcile retention relay).*" } > 26*3600 (relay: > 900 s)</code>	—	ticket	SOP-09
ALR-36	ReconciliationMismatch	<code>Log/SQL: last reconciliation job result mismatches > 0 (from GET /api/ops/jobs, or tasco_db query on job_runs.data->'result'->'mismatches')</code>	—	ticket (page if > 10)	RB-11

6.4 Security and audit

ID	Name	Expression	For	Severity	Runbook
ALR-40	AuditChainBroken	<code>Synthetic SYN-07 result ok == false (exported as probe_success{probe="audit_verify"} == 0)</code>	—	page	RB-08
ALR-50	LoginFailureSpike	<code>sum(increase(http_requests_total{route="POST /api/auth/login",status="401"}[5m])) > 30</code>	—	ticket (page if > 200)	RB-09
ALR-51	AccountLockouts	<code>sum(increase(http_requests_total{route="POST /api/auth/login",status="423"}[15m])) > 5</code>	—	ticket	RB-09
ALR-52	LoginRateLimited	<code>sum(increase(http_requests_total{route=~"POST /api/auth/(login mfa)",status="429"}[5m])) > 20</code>	—	ticket	RB-09
ALR-53	MfaFailures (log/audit)	<code>Audit auth.mfa_failed count > 20 in 15 min (SQL exporter on audit_log.action)</code>	—	ticket	RB-09
ALR-54	PartnerAuthFailures	<code>sum(increase(http_requests_total{route=~".*/api/partner/v1/.*",status="401"}[10m])) > 50</code>	—	ticket	RB-15
ALR-55	ForbiddenSpike	<code>sum by (route) (increase(http_requests_total{status="403"}[10m])) > 50 (possible IDOR probing)</code>	—	ticket	Security
ALR-56	DemoModeInProduction	<code>Synthetic GET /api/meta body demoMode:true, or GET /api/demo/totp/admin ≠ 404</code>	—	page	Config, KI-01
ALR-57	InMemoryStoreInProduction	<code>Synthetic GET /health/ready body store != "postgres"</code>	—	page	KI-28

6.5 Resources

ID	Name	Expression	For	Severity	Runbook
ALR-60	MemoryHigh	<code>max(process_resident_memory_bytes) > 0.8 * 1.5 * 1024^3 (adjust to the container limit)</code>	15m	ticket	RB-13
ALR-61	Frequent Restarts	<code>sum(resets(process_uptime_seconds[1h])) > 3</code>	—	ticket	RB-13

ID	Name	Expression	For	Severity	Runbook
AL-R-62	CPUThrottling	<code>sum(rate(container_cpu_cfs_throttled_periods_total{container="app"}[5m])) / sum(rate(container_cpu_cfs_periods_total{container="app"}[5m])) > 0.25</code>	15m	ticket	RB-12

6.6 Business KPIs

ID	Name	Expression	For	Severity	Notes
AL-R-80	NoOrdersInBusinessHours	<code>sum(increase(orders_completed_total[1h])) == 0 and on() (hour() >= 2 and hour() < 13)</code>	—	page (after pilot full volume)	09:00–20:00 ICT
AL-R-81	ConversionDrop	<code>sum(increase(orders_completed_total[6h])) / sum(increase(quotes_total[6h])) < 0.5 * (sum(increase(orders_completed_total[6h] offset 1w)) / sum(increase(quotes_total[6h] offset 1w)))</code>	—	ticket	Week-over-week
AL-R-82	QuotesDropByChannel	<code>sum by (channel) (increase(quotes_total[1h])) < 0.3 * sum by (channel) (increase(quotes_total[1h] offset 1w))</code>	—	ticket	Channel outage
AL-R-83	HotHandoffsUnworked	<code>tasco_db_handoffs_open_older_than_2h > 20 and on() (hour() >= 1 and hour() < 11)</code>	30m	ticket to the telesales supervisor	KPI: first contact ≤ 2 business hours
AL-R-84	Issuance Failures	<code>tasco_db_orders{status="issuance_failed_refunded"} > 0</code>	—	page	RB-11
AL-R-30	CopyGuardBlocked	<code>sum(increase(messages_total{status="blocked"}[1h])) > 0</code>	—	page Compliance + L2	An active template contains banned wording, which validation should prevent: investigate the rule set (RB-10)
AL-R-31	Message Failure Ratio	<code>sum by (channel) (increase(messages_total{status="failed"}[30m])) / sum by (channel) (increase(messages_total[30m])) > 0.1</code>	—	ticket	RB-04
AL-R-85	OptOutSpoke	<code>sum(increase(voice_calls_total{outcome="opted_out"}[2h])) / sum(increase(voice_calls_total[2h])) > 0.1</code>	—	ticket to Compliance	Script or targeting problem
AL-R-86	PlateMismatchSpoke	<code>sum(increase(voice_calls_total{outcome=~"plate_mismatch unverified"}[2h])) / sum(increase(voice_calls_total[2h])) > 0.15</code>	—	ticket	Data quality, or ASR/KI-25

6.7 Database

ID	Name	Expression (postgres_exporter)	For	Severity
ALR-90	DBConnections High	<code>sum(pg_stat_activity_count) / max(pg_settings_max_connections) > 0.8</code>	5m	ticket

ID	Name	Expression (postgres_exporter)	For	Severity
ALR-91	DBReplicationLag	pg_replication_lag_seconds > 30	5m	page (RPO at risk)
ALR-92	DBDiskHigh	DB volume used > 70 % (provider metric)	15m	ticket
ALR-93	BackupFailed	Provider backup/PITR status ≠ success in 24 h	—	page

Example Alertmanager rule file entry:

```
- alert: CircuitOpen
  expr: sum by (integration) (increase(integration_short_circuit_total[5m])) > 0
  for: 1m
  labels:
    severity: '{{ if or (eq $labels.integration "vetc-wallet") (eq $labels.integration "tasco-core") }}page{{ else }}ticket{{ end }}'
  annotations:
    summary: "Circuit open for {{ $labels.integration }}"
    runbook_url: "docs/operations/runbook-and-support-guide.pdf#rb-02--circuit-open-vetc-wallet-vetc-wallet"
```

(Alertmanager cannot template labels in the rule itself on every version. If yours cannot, split the rule into a `page` rule for `integration=~"vetc-wallet|tasco-core"` and a `ticket` rule for the rest.)

7. Log-based alerts

ID	Query (LogQL)	Threshold	Severity
LOG-01	<code>sum(count_over_time({app="tasco-growth-platform"} json msg="unhandled error" [10m]))</code>	> 5	ticket
LOG-02	<code>count_over_time({app="tasco-growth-platform"} json msg="circuit opened" [5m])</code>	> 0	follows ALR-10
LOG-03	<code>count_over_time({app="tasco-growth-platform"} json msg="job failed" [1h])</code>	> 0	ticket
LOG-04	<code>count_over_time({app="tasco-growth-platform"} json msg="pg pool error" [5m])</code>	> 3	page
LOG-05	<code>count_over_time({app="tasco-growth-platform"} json level="warn" ~ "not set – using an ephemeral"[1h])</code>	> 0 in PROD	page (missing secret in a non-production config that reached prod)
LOG-06	Sign-in failures: <code>sum(count_over_time({app="tasco-growth-platform"} json route="POST /api/auth/login" status="401" [5m]))</code> . The audit action <code>auth.login_failed</code> is in the DB, not the logs; use the HTTP 401 on the login route as the log proxy	> 30	ticket
LOG-07	<code>{app="tasco-growth-platform"} ~ "(?i)(\\b0[35789]\\d{8}\\b)"</code> (unredacted VN mobile pattern in logs)	> 0	ticket to Security (redaction gap)

8. Synthetic checks

ID	Check	Frequency / location	Assertion	Alert
SY N-01	GET /health/ready through the public ingress	1 min, 2 locations (HN, HCM)	200, db:true, store:"postgres"	ALR-01, ALR-57
SY N-02	GET /api/public/certificates/<synthetic cert> (a long-dated internal test policy agreed with TASCO core)	5 min	valid:true, masked plate	ALR-71
SY N-03	Staff login with a no-MFA synthetic user (role executive, read-only) → GET /api/auth/me	5 min	200 within 1 s	ticket
SY N-04	Customer flow (session → home → quote) in PREPROD only . In PROD a quote creates real records, so it runs read-only (session + home) for a synthetic test profile excluded from KPIs	10 min	200s within SLO	ticket
SY N-05	GET /api/meta	5 min	demoMode:false	ALR-56
SY N-06	GET /api/demo/totp/admin	1 h	404	ALR-56
SY N-07	GET /api/audit/verify with an auditor service account	1 h, off-peak heavy (KI-12)	ok:true	ALR-40
SY N-08	TLS certificate expiry for the public hostnames	Daily	> 21 days	ticket
SY N-09	/verify/<certNo> page renders (HTML 200)	15 min	200, contains the cert number	ALR-71
AL R-71	Certificate verification synthetic failing	SYN-02 or SYN-09 failing for 10 min	—	page (police/inspection depend on it)

9. Alert hygiene

- Every alert has an owner, a runbook link and a tested expression (unit-tested with `promtool test rules`).
- Silences must have an expiry and a ticket ID.
- Monthly review: alerts that fired with no action are tuned or removed; incidents with no alert get a new rule.
- Thresholds marked for the pilot are re-baselined after pilot week 2 and before W1 (volumes ×20–30).